The Evolution of Information Retrieval:
From Sparse to Neural and Hybrid Retrieval Strategies

A Comparative Empirical Study on BEIR Benchmarks

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Abstract

Information retrieval (IR) has undergone a paradigm shift over the past decade, moving from hand-crafted statistical methods toward deep neural representations and hybrid multi-stage pipelines. This study traces that evolution across four phases: (1) classical sparse retrieval (BoW, TF-IDF, BM25); (2) early neural retrieval (Word2Vec mean/IDF pooling); (3) dense bi-encoder retrieval (MiniLM, MPNet, BGE, E5); (4) hybrid retrieval with cross-encoder reranking (Reciprocal Rank Fusion + `ms-marco-MiniLM-L-6-v2`). We further investigate domain-specific fine-tuning of a 22M-parameter encoder using MultipleNegativesRankingLoss.

All experiments are evaluated on three BEIR benchmarks with distinct characteristics: FIQA (financial Q&A, conversational), SciFact (scientific claim verification), and TREC-COVID (biomedical passage retrieval). Our results show that BGE-base dominates zero-shot dense retrieval (NDCG@10 0.78 on TREC-COVID); the cross-encoder reranker is a more reliable upgrade than hybrid RRF when the sparse component is weak: BGE alone (0.781) outperforms hybrid+reranker (0.763) on TREC-COVID; and 919-pair domain-specific fine-tuning yields +7.7% in-domain NDCG with minimal negative transfer across other domains. We identify concrete failure modes at each stage: BM25 length-normalisation degrading on uniform-length corpora, Word2Vec silently dropping out-of-vocabulary biomedical identifiers, DPR collapsing on out-of-domain queries, and equal-weight RRF diluting a dominant dense signal.

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1 Introduction

Information retrieval underpins almost every modern NLP application, from search engines to retrieval-augmented generation (RAG) systems. Historically, term-overlap methods such as BM25 (Robertson and Zaragoza, 2009) dominated the field, offering strong baselines with no training requirements. The introduction of dense embeddings via BERT (Devlin et al., 2019) and the Sentence-BERT (SBERT) bi-encoder framework (Reimers and Gurevych, 2019) opened a new era where semantic similarity could be computed in a continuous vector space, enabling retrieval of relevant documents even when query and passage share no lexical overlap. SBERT established the Siamese bi-encoder architecture that all subsequent dense retrievers build upon; this study evaluates its successors (MiniLM, MPNet, BGE, E5) rather than the original model.

More recently, instruction-tuned embedding models such as BGE (Xiao et al., 2023) and E5 (Wang et al., 2022) have pushed the frontier further, while hybrid pipelines that fuse sparse and dense signals via Reciprocal Rank Fusion (RRF) (Cormack et al., 2009) followed by cross-encoder reranking have become the de-facto production architecture. Simultaneously, large language models (LLMs) are increasingly used as *judges* to evaluate retrieval quality beyond the limitations of manually annotated relevance judgements (qrels).

Contributions.

1. A systematic, reproducible comparison of four IR paradigms on three heterogeneous BEIR benchmarks using unified evaluation metrics.
2. Identification of concrete failure modes at each stage: BM25 length-normalisation degrading on uniform-length corpora (SciFact, FIQA); Word2Vec silently dropping OOV biomedical identifiers; DPR collapsing out-of-domain; equal-weight RRF diluting a dominant dense signal when the sparse component is weak.
3. Per-query working and failure examples for each method, making failure modes tangible rather than aggregate.
4. Empirical evidence that 919-pair domain fine-tuning with MultipleNegativesRankingLoss yields competitive gains, and that combined multi-domain training generalises to unseen domains without negative transfer.

2 Background

2.1 Evaluation Benchmarks

We evaluate on three BEIR (Thakur et al., 2021) datasets spanning diverse retrieval domains and query types. Table 1 summarises the key statistics.

Table 1: BEIR benchmark datasets used in this study.

Dataset	Domain	Task	Corpus	Test queries	Relevance	Avg. words/passage
FIQA	Financial Q&A	Answer retrieval	57,638	648	Binary (0/1)	133
SciFact	Scientific claims	Claim verification	5,183	300	Binary (0/1)	215
TREC-COVID	Biomedical	Passage retrieval	171,332	50	Graded (0/1/2)	161

FIQA 648 financial community Q&A questions; 57,638 passages. Queries are informal, conversational, and domain-specific (sourced from Reddit and StackExchange). Relevance labels in the original FiQA challenge were graded; the BEIR test split binarises all qrels to score = 1. BM25 is well-known to underperform on conversational queries.

SciFact 300 scientific claim queries; 5,183 abstracts. Relevance requires domain understanding; exact-term matching often fails on paraphrase-style claims. On average only 1.1 relevant passages exist per query, making Recall@10 a demanding metric.

TREC-COVID 50 COVID-19 biomedical questions; 171,332 PubMed abstracts. Relevance is graded: 0 (not relevant), 1 (partially), 2 (highly). With 493 relevant passages per query on average, Recall@100 captures only ∼12% of all relevant documents (a corpus-coverage limit, not a model limit).

Example queries and passages. To illustrate the diversity of query styles across datasets:

- **FIQA**: *"How should I invest my first $1,000?"*
  Relevant passage: "For small amounts, low-cost index funds tracking the S&P 500 provide broad diversification with minimal fees. Consider a Roth IRA if you have earned income..."
- **SciFact**: *"Headaches are not correlated with cognitive impairment."*
  Relevant passage: "Analysis of 412 patients found no significant association between cephalgia frequency and neurocognitive function scores (p = 0.43), after adjusting for age and comorbidities."
- **TREC-COVID**: *"What are the symptoms of COVID-19?"* (graded relevance; same query, three passages at different relevance levels)
  - **Grade 2 (highly relevant):** "Common manifestations include fever, dry cough, and fatigue. Less common symptoms are myalgia, sore throat, diarrhoea, conjunctivitis, and headache. Dyspnoea occurs in severe cases."
  - **Grade 1 (partially relevant):** "Coronavirus infections generally present with respiratory symptoms; some patients remain asymptomatic with detectable viral load, complicating containment."
  - **Grade 0 (not relevant):** "Influenza vaccination rates among healthcare workers declined during the 2019–20 season due to temporary supply shortages at regional distribution centres."

Metric validity on graded relevance (TREC-COVID). TREC-COVID uses graded relevance (0, 1, 2), while FIQA and SciFact use binary labels. NDCG@10 handles graded relevance natively via $\text{rel}_i / \log_2(i+1)$ and is the primary metric throughout this study. MRR and Recall, however, binarise at grade ≥ 1: MRR treats a partially relevant (grade 1) hit identically to a highly relevant (grade 2) hit, and Recall@k counts all grade-≥ 1 documents as the relevant set, which includes 493 passages per query on average in TREC-COVID. These two effects make MRR and Recall less informative for TREC-COVID than

for the other two datasets; NDCG@10 should be the basis for any cross-dataset comparison involving TREC-COVID.

Note the vocabulary gap in the SciFact example: the query uses "headaches" while the passage uses "cephalgia". Lexical methods score zero; vector-based methods can bridge this gap when the words are nearby in embedding space.

2.2 Evaluation Metrics

Following BEIR convention we report the metrics below. All metrics are averaged over queries.

NDCG@10 (Normalised Discounted Cumulative Gain). Primary ranking quality metric. Rewards retrieving relevant documents early in the ranking.

$$\mathrm{DCG@}k = \sum_{i=1}^{k} \frac{\mathrm{rel}_i}{\log_2(i+1)}, \qquad \mathrm{NDCG@}k = \frac{\mathrm{DCG@}k}{\mathrm{IDCG@}k} \tag{1}$$

where $\mathrm{rel}_i \in \{0, 1, 2\}$ is the relevance of the document at rank i, and IDCG@k is the ideal (best-possible) DCG@k.

MRR@10 (Mean Reciprocal Rank). Measures how quickly the first relevant document is found.

$$\mathrm{MRR@}k = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\mathrm{rank}_q^{\text{first relevant}}} \tag{2}$$

The retrieved list is truncated to top-k *before* passing to `pytrec_eval`, preventing rank-cutoff inflation on weaker models.

MAP@100 (Mean Average Precision).

$$\mathrm{MAP@100} = \frac{1}{|Q|} \sum_{q \in Q} \mathrm{AP}_q, \qquad \mathrm{AP}_q = \frac{1}{R_q} \sum_{k=1}^{100} P@k \cdot \mathbb{1}[\mathrm{doc}_k \text{ relevant}] \tag{3}$$

where R_q is the total number of relevant documents for query q, and $P@k$ is precision at rank k.

Recall@k.

$$\mathrm{Recall@}k = \frac{|\mathrm{Rel} \cap \mathrm{Retrieved@}k|}{|\mathrm{Rel}|} \tag{4}$$

Reported at $k \in \{10, 50, 100\}$. On TREC-COVID, Recall@100 is inherently low ($\sim$0.12) due to the large number of relevant documents per query.

2.3 Reciprocal Rank Fusion

Given ranked lists $L_1, \ldots, L_n$ from n retrievers, the RRF score for document d is:

$$\mathrm{RRF}(d) = \sum_{i=1}^{n} \frac{1}{k + \mathrm{rank}_i(d)} \tag{5}$$

where $k = 60$ (standard default) smooths the contribution of documents ranked low in one list but high in another, preventing a single high-rank appearance from dominating the fusion. We fuse BM25 and BGE ranked lists.

2.4 Training Loss Functions

Three loss functions are widely used for training dense retrievers.

Contrastive Loss (Hadsell et al., 2006). The earliest metric-learning loss. Given a labelled pair (x_i, x_j) with label $y \in \{0, 1\}$ (1 = similar, 0 = dissimilar) and distance $d = \|f(x_i) - f(x_j)\|$:

$$\mathcal{L}_{\text{contrastive}} = y \cdot d^2 + (1 - y) \cdot \max(0,\, m - d)^2 \tag{6}$$

where $m > 0$ is a margin. Requires explicit pair labels; does not scale well to large corpora.

Triplet Loss (Schroff et al., 2015). Trains on (query, positive, negative) triplets. Used in the original SBERT:

$$\mathcal{L}_{\text{triplet}} = \max\big(0,\ d(q, p^+) - d(q, p^-) + m\big) \tag{7}$$

where d is a distance function (e.g. cosine or Euclidean), p^+ is a relevant passage, p^- is a hard negative, and m is the margin (typically 0.5–1.0). Requires explicit hard-negative mining.

MultipleNegativesRankingLoss (MNR / InfoNCE). Current best practice for bi-encoder training (Henderson et al., 2017):

$$\mathcal{L}_{\text{MNR}} = -\frac{1}{B} \sum_{i=1}^{B} \log \frac{e^{\,\text{sim}(\mathbf{q}_i,\, \mathbf{p}_i)/\tau}}{\sum_{j=1}^{B} e^{\,\text{sim}(\mathbf{q}_i,\, \mathbf{p}_j)/\tau}} \tag{8}$$

where B is the batch size, $(\mathbf{q}_i, \mathbf{p}_i)$ is the i-th (query, positive passage) pair, and τ is the temperature. All other positives in the batch act as *in-batch negatives*; no hard-negative mining is required. We use MNR in this study because BEIR qrels directly supply (query, positive) pairs and batch mates are free negatives.

Table 2: Comparison of bi-encoder training losses.

Loss	**Negatives**	**Scalability**	**Notes**
Contrastive	Explicit labelled pairs	Low	Early metric learning
Triplet	Hard-mined triplets	Medium	SBERT v1; needs mining
MNR/InfoNCE	In-batch (free)	High	Current best practice

3 Experimental Setup

Hardware Apple M5 (15-core CPU, 16-core GPU, 24 GB unified memory). PyTorch MPS backend; FAISS replaced with exact numpy dot-product search to avoid OpenMP library conflicts on macOS (`libomp` conflict #15 / SIGABRT).

Software Python 3.12; `sentence-transformers 5.4.1`; `transformers 4.x`; `torch 2.x` (MPS backend); `gensim 4.x` (Word2Vec); `scikit-learn` (TF-IDF/BoW); `rank-bm25`; `beir`; `pytrec_eval`.

Replication All scripts in `scripts/`; results persisted to `results/` as JSON; notebooks in `notebooks/` reproduce all plots and tables.

4 Phase 1: Sparse Retrieval

4.1 Methods

Bag-of-Words (BoW)

Each document is represented as a vector of raw term counts over the corpus vocabulary V. Query–document similarity is cosine similarity over count vectors:

$$\text{score}_{\text{BoW}}(q, d) = \cos(\mathbf{v}_q,\, \mathbf{v}_d), \quad \mathbf{v}_d[t] = \text{tf}(t, d) \tag{9}$$

where $\text{tf}(t, d)$ is the **term frequency**, i.e., the raw count of how many times term t appears in document d. **Limitation:** high-frequency stop words ("the", "is", "a") dominate the dot product and dilute discriminative signal.

TF-IDF

Weights each term by its *inverse document frequency*, down-weighting common words and up-weighting rare, discriminative ones:

$$w(t, d) = \text{tf}(t, d) \times \text{idf}(t), \quad \text{idf}(t) = \log \frac{N}{\text{df}(t)} \tag{10}$$

$$\text{score}_{\text{TF-IDF}}(q, d) = \cos(\mathbf{w}_q,\, \mathbf{w}_d) \tag{11}$$

where N is the corpus size and $\text{df}(t)$ is the number of documents containing term t. In practice, scikit-learn applies sublinear TF scaling ($\text{tf} \leftarrow 1 + \log \text{tf}$), dampening the effect of high-frequency terms. The cosine similarity implicitly L2-normalises document vectors, providing a soft global length normalisation.

BM25 (Best Match 25)

BM25, introduced in the early 1990s (Robertson et al., 1994) and formally surveyed in Robertson and Zaragoza (2009), adds two important corrections over TF-IDF:

$$\text{score}_{\text{BM25}}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, d) \cdot (k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b\, \dfrac{|d|}{\text{avgdl}}\right)} \tag{12}$$

$$\text{IDF}(t) = \log \left(\frac{N - \text{df}(t) + 0.5}{\text{df}(t) + 0.5} + 1 \right) \tag{13}$$

Parameters:

- $f(t, d)$: term frequency of term t in document d (identical to $\text{tf}(t, d)$ in Eq. 9). The saturation formula in the numerator/denominator replaces the raw count used in TF-IDF.

- $k_1 \in [1.2, 2.0]$: *term-frequency saturation*: controls how much weight additional occurrences of a term add. At $k_1 = 0$ all occurrences beyond the first are ignored; as $k_1 \to \infty$ the score approaches raw TF. Default: 1.5.
- $b \in [0, 1]$: *length normalisation strength*: $b = 0$ disables length penalty, $b = 1$ fully normalises by document length. Default: 0.75 (calibrated on MS MARCO and similar long-document collections).
- $|d|$: document length in terms; avgdl: corpus-average length.

4.2 Experimental Results

Table 3: Sparse retrieval: NDCG@10 across all datasets.

Method	TREC-COVID	FIQA	SciFact
BoW	0.1763	0.0656	0.3647
TF-IDF	0.2866	**0.1793**	**0.6286**
BM25	**0.4471**	0.1591	0.5597

Findings. BM25 is the best sparse method on TREC-COVID, but TF-IDF beats BM25 on FIQA and SciFact. The gap on TREC-COVID (0.287 vs. 0.447) illustrates BM25's length-normalisation advantage on long biomedical abstracts (mean 161 words, high variance). The TF-IDF inversion on FIQA and SciFact is *structural*, not a hyperparameter issue, as detailed in Section 4.3 below.

Illustrative per-query examples.

- **BoW worked.**
  Query: "AIRE is expressed in some skin tumors." (SciFact)
  Expected passage: Keratin-dependent regulation of Aire and gene expression in skin tumor keratinocytes. "Expression of K17 is robustly upregulated in tumors originating in stratified epithelia; Aire is inducibly expressed in human and mouse tumor keratinocytes."
  Why it worked: Three rare tokens (`AIRE`, `skin`, `tumors`) appear verbatim in the passage; their high cosine contribution puts the relevant passage at rank 1 despite no stop-word suppression.
- **BoW failed (zero vocabulary overlap).**
  Query: "0-dimensional biomaterials show inductive properties." (SciFact)
  Expected passage: New opportunities: the use of nanotechnologies to manipulate and track stem cells. "Examples include magnetic nanoparticles and quantum dots for stem cell labeling and in vivo tracking."
  Retrieved (rank 1): US28 chemokine receptor constitutive endocytosis. "US28 is one of four transmembrane chemokine receptors encoded by human cytomegalovirus."
  Why it failed: No token in the query (`0-dimensional`, `biomaterials`, `inductive`) overlaps with any token in the relevant passage (`nanotechnologies`, `quantum`, `dots`); cosine similarity is zero and stop words push a completely unrelated CMV passage to rank 1.

- **TF-IDF won where BoW failed.**
  Query: "ADAR1 binds to Dicer to cleave pre-miRNA." (SciFact)
  Expected passage: ADAR1 Forms a Complex with Dicer to Promote MicroRNA Processing. "ADAR1 forms a complex with Dicer through direct protein-protein interaction and increases the maximum rate of pre-miRNA cleavage by Dicer."
  BoW rank-1 (wrong): Phosphate and R2D2 restrict the substrate specificity of Dicer-2. "Drosophila Dicer-2 generates siRNAs from long double-stranded RNA; Dicer-1 produces miRNAs from pre-miRNA." (`Dicer` matches, but `ADAR1` is absent)
  Why TF-IDF won: `ADAR1` is rare across the corpus and carries very high IDF; TF-IDF amplifies its weight to lift the correct passage to rank 1. BoW treats `Dicer` and `ADAR1` as equally weighted and retrieves a Dicer-only passage.

- **All sparse methods failed (surface overlap cannot resolve claim specificity).**
  Query: "A deficiency of vitamin B12 increases blood levels of homocysteine." (SciFact)
  Expected passage: Randomized trial of folic acid supplementation and serum homocysteine levels. "Lowering serum homocysteine with folic acid is expected to reduce mortality from ischemic heart disease; homocysteine reduction is maximal at 1 mg/d folic acid."
  Retrieved (rank 1, all three methods): BoW, TF-IDF, and BM25 each retrieve a different passage from the same metabolic topic area, but none retrieves the annotated relevant passage at rank 1. For example, BM25 returns: *Folate status and plasma homocysteine response to supplementation.* "High plasma total homocysteine (tHcy) is a risk factor for vascular diseases; we investigated the extent to which serum folate and plasma tHcy respond to supplementation."
  Why all failed: The query, the expected passage, and the retrieved passages all share surface tokens (`homocysteine`, `vitamin`, `blood`, `levels`, `deficiency`, `folate`). No lexical method, regardless of weighting scheme or length normalisation, can distinguish which passage corroborates the specific B12→homocysteine causal claim versus discussing the same metabolic pathway from a different angle. This is a structural limit of surface-form matching, not a tuning issue.

4.3 BM25 Hyperparameter Sweep

To investigate why TF-IDF beats BM25 on SciFact and FIQA, we sweep the length-normalisation parameter b and the saturation parameter k_1. Table 4 shows corpus passage-length statistics.

Table 4: Passage-length statistics per dataset (words). CV = coefficient of variation (σ/μ): a dimensionless measure of length dispersion. Higher CV indicates more variable passage lengths, making length normalisation more important.

Dataset	Count	Mean	Median	Std	CV
SciFact	5,183	214.6	204.0	87.3	0.41
FIQA	57,638	132.9	90.0	128.7	0.97
TREC-COVID	171,332	161.2	168.0	135.0	0.84

Table 5: b sweep ($k_1 = 1.5$ fixed): NDCG@10. Bold = best BM25 per dataset. TF-IDF column for reference.

Dataset	$b = 0.0$	$b = 0.25$	$b = 0.5$	$b = 0.75$	$b = 1.0$	**TF-IDF**
SciFact	0.547	0.558	0.556	0.560	**0.563**	**0.629**
FIQA	0.099	0.147	**0.162**	0.159	0.039	**0.179**
TREC-COVID	0.337	0.460	**0.479**	0.447	0.317	0.287

Table 6: k_1 sweep ($b = 0.75$ fixed): NDCG@10. Bold = best BM25 per dataset.

Dataset	$k_1 = 0.5$	$k_1 = 1.0$	$k_1 = 1.5$	$k_1 = 2.0$	$k_1 = 2.5$
SciFact	0.559	0.563	0.560	0.562	**0.567**
FIQA	0.162	**0.163**	0.159	0.156	0.151
TREC-COVID	0.424	0.437	**0.447**	0.438	0.441

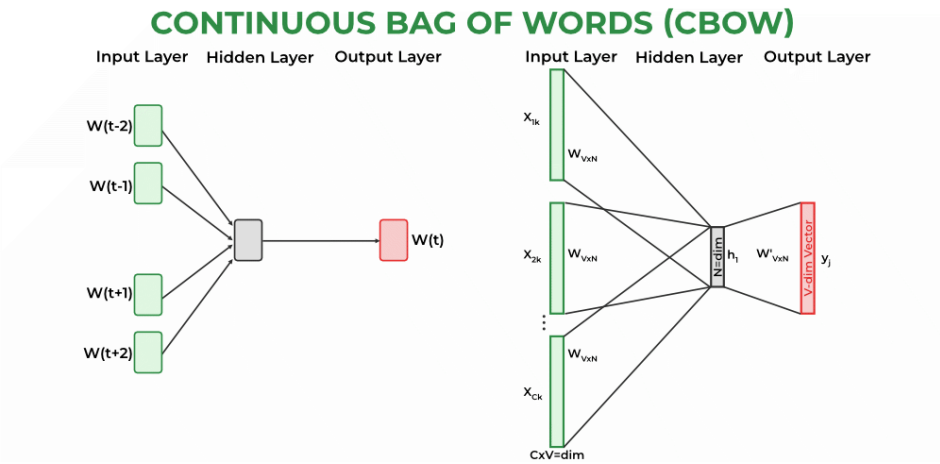
Interpretation. The TF-IDF advantage on SciFact and FIQA is *structural*: even setting $b = 0$ (completely disabling length normalisation) does not close the gap. The root cause is that TF-IDF's cosine similarity provides a *global, implicit* L2 normalisation that outperforms BM25's *explicit, per-term* length formula on short, relatively uniform passages. BM25's b parameter was calibrated on MS MARCO (long, highly variable documents); it does not transfer well to SciFact (CV 0.41) or FIQA. The best BM25 on TREC-COVID is $b = 0.5$, confirming that the default $b = 0.75$ over-normalises medium-length biomedical abstracts.

5 Phase 2: Early Neural Retrieval

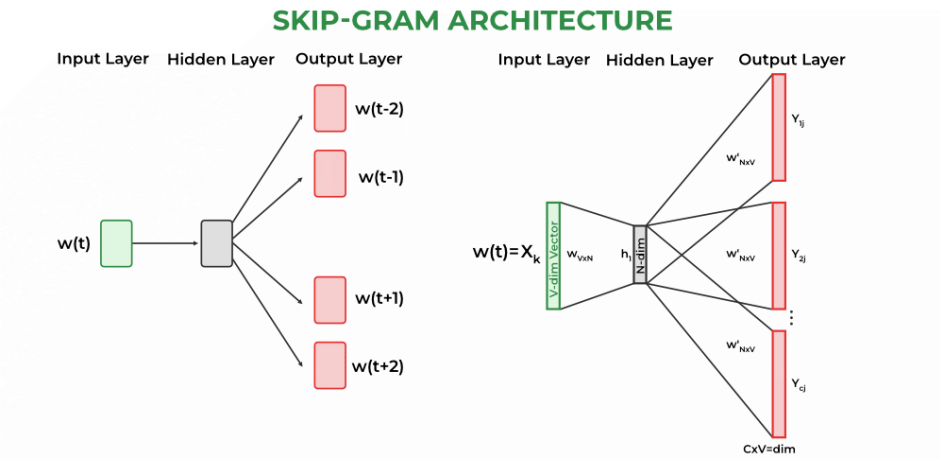
5.1 Word2Vec Architecture

Word2Vec (Mikolov et al., 2013) trains a shallow neural network on a large text corpus (Google News, ∼100B words) to produce 300-dimensional word vectors. Two complementary architectures are used:

- **Skip-gram:** Given a centre word w_t, predict the surrounding context words w_{t+j} within a window of size W. Objective: maximise $\sum_{j \neq 0, |j| \leq W} \log P(w_{t+j} \mid w_t)$. Better at representing rare words; used for the Google News vectors.
- **CBOW (Continuous Bag-of-Words):** Given the surrounding context words, predict the centre word. Faster to train; slightly weaker on rare vocabulary.



(a) CBOW: context words (input) → centre word (predicted).



(b) Skip-gram: centre word (input) → surrounding context words (predicted).

Figure 1: Word2Vec architectures. CBOW averages context embeddings to predict the centre word; Skip-gram inverts this task, predicting all context words from the centre word. Skip-gram is slower but learns better representations for rare vocabulary; the Google News checkpoint uses Skip-gram. Source: GeeksForGeeks (`https://www.geeksforgeeks.org/nlp/word-embeddings-in-nlp-comparison-between-cbow-and-skip-gram-models/`); architecture from Mikolov et al. (2013).

After training, semantically similar words cluster nearby in the 300-d space: $\text{vec}(\text{king}) - \text{vec}(\text{man}) + \text{vec}(\text{woman}) \approx \text{vec}(\text{queen})$.

Polysemy collapse. A key limitation: every word has exactly one vector regardless of context. The word "bank" receives the same vector whether used in a financial query ("savings bank") or a geographical one ("river bank"). This is a fundamental flaw for retrieval, where the same surface form may match irrelevant passages.

GloVe: Global Vectors (Pennington et al., 2014). Published the same year as Word2Vec, GloVe (Global Vectors for Word Representation) takes a different training objective but produces similar 300-d static word vectors. Rather than predicting neigh-

bouring words from a centre word (a *local* task); GloVe factorises the global word–word co-occurrence matrix built from the entire corpus:

$$J = \sum_{i,j=1}^{V} f(X_{ij})\big(\mathbf{w}_i^\top \tilde{\mathbf{w}}_j + b_i + \tilde{b}_j - \log X_{ij}\big)^2 \tag{14}$$

where X_{ij} is the co-occurrence count of words i and j, $f(\cdot)$ is a weighting function that down-weights very frequent pairs, and $\mathbf{w}_i, \tilde{\mathbf{w}}_j$ are the word and context vectors being learned. The intuition is that the *ratio* of co-occurrence probabilities carries meaning: $P(\text{ice}|\text{solid})/P(\text{steam}|\text{solid}) \gg 1$, differentiating "solid" from related but distinct concepts.

Table 7: Word2Vec vs. GloVe: key differences.

Property	**Word2Vec (Skip-gram)**	**GloVe**
Training signal	Local window prediction	Global co-occurrence matrix
Objective	Cross-entropy (negative sampling)	Weighted least squares
Corpus statistics	Implicit, sampled	Explicit, precomputed
Typical dim.	300	50, 100, 200, 300
Common checkpoint	Google News (100B words)	Wikipedia + Gigaword (6B)
Analogy tasks	Comparable	Slightly stronger

In practice, GloVe and Word2Vec perform comparably on most IR benchmarks; the choice between them is usually driven by the available pre-trained checkpoint rather than any intrinsic superiority. Both share the same fundamental limitation: **static, context-independent representations** that cannot resolve polysemy. Our experiments use the Google News Word2Vec checkpoint; GloVe is presented here as the natural complement and to complete the picture of the static-embedding era before contextual models.

Related static-embedding variants (not evaluated, shortlisted). Several extensions address known Word2Vec limitations and are shortlisted for future experimentation:

- **FastText** (Bojanowski et al., 2017): represents each word as a sum of character n-gram vectors, eliminating OOV entirely. Terms like `PCNA` or `cSMAC`, absent from Google News, are decomposed into subword pieces, giving non-zero vectors. Drop-in replacement for Word2Vec in the pooling pipeline.
- **Doc2Vec / Paragraph Vectors** (Le and Mikolov, 2014): trains a dedicated document-level vector $\mathbf{d}_i \in \mathbb{R}^{300}$ alongside word vectors. Two variants: PV-DM (document ID as an extra context slot) and PV-DBOW (document ID predicts sampled words). Unlike pooling, the document vector is a *trained* parameter; it captures passage-level semantics without the long-passage centroid collapse that hurts mean and IDF-weighted pooling on FIQA and SciFact. PV-DBOW is shortlisted as the primary experiment; implementation planned in `scripts/run_doc2vec_eval.py`.

5.2 Methods: Pooling Strategies

Both aggregation strategies represent a passage as a single vector by combining its token vectors.

Mean Pooling. Uniform average over all token vectors in the passage:

$$\mathbf{d}_{\text{mean}} = \frac{1}{|T|} \sum_{t \in T} \mathbf{v}_t \tag{15}$$

where T is the sequence of tokens and $\mathbf{v}_t \in \mathbb{R}^{300}$ is the Word2Vec vector for token t. Equal weight is given to all tokens, including stop words, which dilutes the semantic signal.

IDF-Weighted Pooling. Weights each token by its IDF score before averaging, suppressing high-frequency stop words and up-weighting rare, discriminative terms:

$$\mathbf{d}_{\text{IDF}} = \frac{\sum_{t \in T} \text{idf}(t) \cdot \mathbf{v}_t}{\sum_{t \in T} \text{idf}(t)}, \quad \text{idf}(t) = \log \frac{N}{\text{df}(t)} \tag{16}$$

This is the same intuition as TF-IDF applied in the vector space rather than the term-frequency space.

5.3 Experimental Results

Table 8: Early neural retrieval: NDCG@10.

Method	TREC-COVID	FIQA	SciFact
Word2Vec (mean)	0.3393	0.0596	0.2687
Word2Vec (IDF-wt.)	**0.4365**	**0.0885**	**0.3101**
BM25 (reference)	0.4471	0.1591	0.5597

Findings.

- **IDF-weighting consistently improves mean pooling.** Gains of +0.097 on TREC-COVID, +0.029 on FIQA, +0.041 on SciFact; suppressing stop words matters even in embedding space.
- **The gap to BM25 tracks domain distance from Google News.** Word2Vec IDF is near-equal to BM25 on TREC-COVID (−0.011) but far below on SciFact (−0.250). Biomedical and scientific vocabularies are underrepresented in Google News, so learned vectors are poor representations for domain-specific terms.
- **Word2Vec wins on paraphrase-style queries, loses on specialised terminology.** Per-query analysis on SciFact shows, for example, Word2Vec beats BM25 by +0.667 NDCG on queries like *"Headaches are not correlated with cognitive impairment"* (common vocabulary, bridgeable synonym gap), but loses by −1.0 on queries containing biomedical identifiers like *"PCNA K164"* or *"cSMAC formation"* (out-of-vocabulary, silently dropped).
- **Long passages hurt averaging disproportionately.** SciFact passages (median 204 words) produce noisier centroid vectors than TREC-COVID passages (median 168 words). Averaging more vectors regresses toward a generic document mean.

- **OOV is a silent failure mode.** Biomedical terms absent from the Google News vocabulary are dropped entirely. BM25 scores an exact lexical match; Word2Vec has no signal at all, losing by a full NDCG point on those queries.

Illustrative per-query examples.

- **Word2Vec won where BM25 failed.**
  Query: "Headaches are not correlated with cognitive impairment." (SciFact)
  Expected passage: Headache, migraine, and structural brain lesions and function: population based study. "The cross sectional study evaluated the association of overall and specific headaches with volume of white matter hyperintensities, brain infarcts, and cognition."
  BM25 rank-1 (wrong): Management of chronic tension-type headache with tricyclic antidepressant medication. "Chronic tension-type headaches are characterized by near-daily headaches and are difficult to manage; behavioral and pharmacological therapies are evaluated." (matches `headache` but is about treatment, not cognitive correlation)
  Why Word2Vec won: The relevant passage uses paraphrase vocabulary (*not correlated* $\leftrightarrow$ *association with cognition*). Word2Vec's embedding space bridges this synonym gap; BM25 latches onto `headache` and retrieves treatment passages instead.

- **Word2Vec failed (silent OOV drop).**
  Query: "Ubiquitin ligase UBC13 generates a K63-linked polyubiquitin moiety at PCNA K164." (SciFact)
  Expected passage: Replication Fork Slowing and Reversal upon DNA Damage Require PCNA Polyubiquitination. "DNA damage tolerance during eukaryotic replication is orchestrated by PCNA ubiquitination; polyubiquitination activates an error-free damage bypass pathway."
  Retrieved (rank 1): 131-I radioiodine therapy for hyperthyroidism in patients with Graves' disease. (completely unrelated; relevant passage appears at rank 5)
  Why it failed: `UBC13`, `K63-linked`, and `PCNA` are absent from Google News and are silently dropped. The surviving generic tokens (`ubiquitin`, `moiety`, `generates`) match thousands of unrelated passages. BM25 and TF-IDF both hit rank 1 via exact lexical match on `PCNA`.

6 Phase 3: Dense Retrieval

6.1 Bi-encoder Architecture

Dense bi-encoders address both key failures of Word2Vec: the OOV problem and polysemy collapse. Two transformer towers (which may share weights) encode query and passage *independently* into fixed-size dense vectors:

$$\mathbf{q} = \text{Encoder}_Q(q) \in \mathbb{R}^d \tag{17}$$

$$\mathbf{p} = \text{Encoder}_P(p) \in \mathbb{R}^d \tag{18}$$

$$\text{score}(q, p) = \mathbf{q} \cdot \mathbf{p} \tag{19}$$

Subword tokenisation (WordPiece / BPE): every term, including rare biomedical identifiers, is decomposed into known subword pieces, eliminating OOV entirely. **Contextual**

representations: unlike Word2Vec, the same surface form receives a different vector depending on its surrounding context.

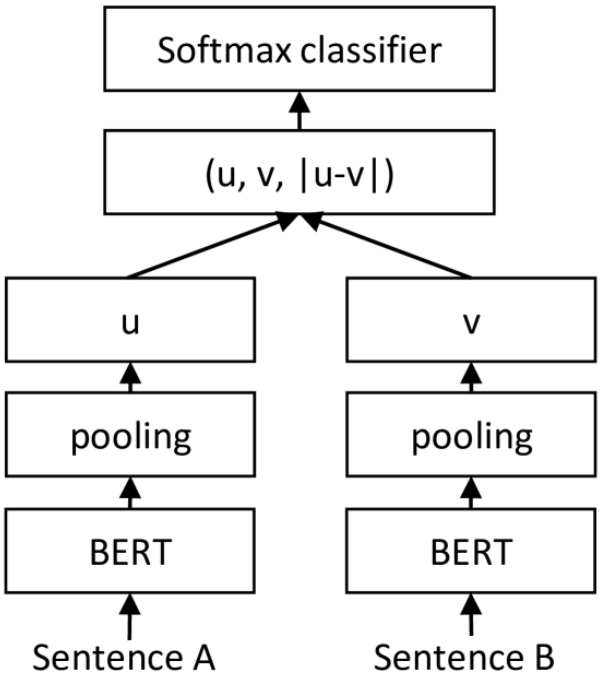


Figure 2: Siamese bi-encoder architecture for dense retrieval, as introduced by SBERT (Reimers and Gurevych, 2019). Query and passage are encoded independently; relevance is computed as a dot product over the resulting vectors. All dense models evaluated in this study (MiniLM, MPNet, BGE, E5) follow this architecture. Source: Reimers and Gurevych (2019) via ar5iv (`https://arxiv.org/abs/1908.10084`).

DPR (Dense Passage Retrieval) (Karpukhin et al., 2020). Uses *separate* question and context encoders, fine-tuned exclusively on Natural Questions (NQ) QA pairs. State-of-the-art on NQ in 2020, but the narrow training distribution causes catastrophic degradation on out-of-domain datasets (see Results, Table 10).

BGE and E5 instruction prefixes. BGE-base-en-v1.5 prepends `"Represent this sentence for searching relevant passages: "` at training time for asymmetric retrieval. E5-base-v2 uses `"query: "` / `"passage: "` prefix tokens to signal the encoding role. These instruction-aware models achieve strong zero-shot performance via large-scale contrastive pre-training with hard negative mining.

6.2 Model Architecture Details

MiniLM: Deep Self-Attention Distillation (Wang et al., 2020). `all-MiniLM-L6-v2` is a 22M-parameter student distilled from a large MPNet teacher using *self-attention transfer*: the student mimics the teacher's self-attention distributions and value-relation matrices, not just the final hidden states. This makes it 5× smaller than its teacher while retaining >99% of retrieval quality on MS MARCO.

Table 9: Dense bi-encoder models and their key references.

Model	Params	Year	Key contribution	ArXiv
DPR (question enc.)	110M	2020	Task-specific dual-encoder on NQ	2004.04906
all-MiniLM-L6-v2	22M	2021	Distillation from MPNet; 1B pairs	N/A
all-mpnet-base-v2	110M	2021	Permuted LM + MLM objective	2004.09297
BGE-base-en-v1.5	110M	2023	Hard-neg mining + instruction	2309.07597
E5-base-v2	110M	2022	Weakly-supervised contrastive pre-training	2212.03533

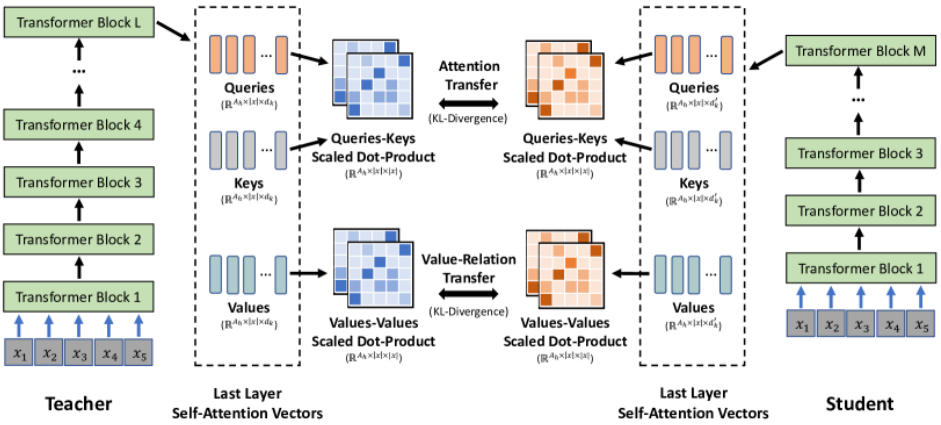


Figure 3: MiniLM: deep self-attention distillation. The student mimics the teacher's self-attention distributions and value-relation matrices layer by layer. Source: Wang et al. (2020) via ar5iv (`https://arxiv.org/abs/2002.10957`).

MPNet: Masked and Permuted Language Modelling (Song et al., 2020). BERT's masked LM (MLM) predicts masked tokens but cannot see their positional context. XLNet's permuted LM (PLM) uses autoregressive prediction but misses dependency between predicted tokens. MPNet combines both: it uses a PLM objective *with* auxiliary position information from the full sentence, eliminating the pretrain–finetune discrepancy.

E5: Weakly-Supervised Contrastive Pre-training (Wang et al., 2022). E5 curates ∼1.3B (text, text$^+$) pairs from web sources (CCPairs) with consistency filtering. Stage 1 is large-scale contrastive pre-training with in-batch negatives on this weakly labelled data. Stage 2 fine-tunes on high-quality labelled data (MS MARCO, NLI). The `"query: "` / `"passage: "` prefix tells the encoder which role to play during asymmetric retrieval.

BGE: Hard-Negative Mining + Instruction Tuning (Xiao et al., 2023). BGE (BAAI General Embedding) trains with a two-step recipe: (1) contrastive pre-training on large web corpora; (2) fine-tuning with *hard negatives* mined from an initial retrieval run. Hard negatives are passages ranked high by BM25 or a weaker dense model but labelled non-relevant, forcing the model to learn fine-grained discrimination. BGE's instruction prefix (`"Represent this sentence for searching relevant passages: "`) signals the encoding role for asymmetric retrieval, similar to E5.

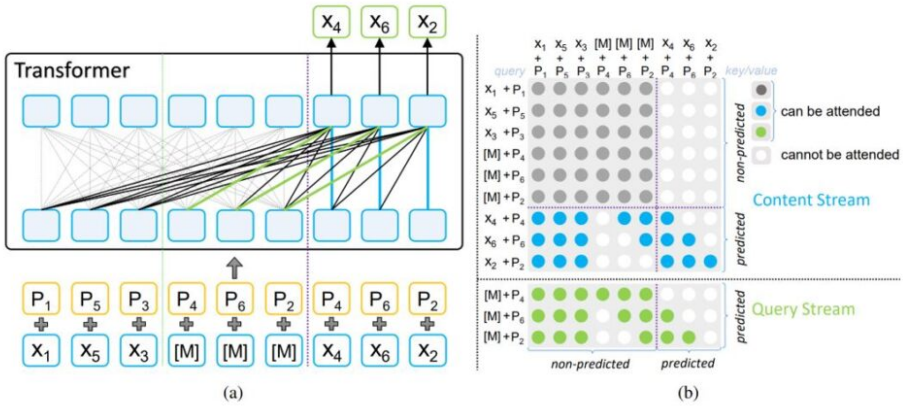


Figure 4: MPNet attention-mask patterns for MLM, PLM, and the combined MPNet objective. Source: Song et al. (2020) via Microsoft Research blog (`https://www.microsoft.com/en-us/research/blog/mpnet-combines-strengths-of-masked-and-permuted-language-modeling-for-language-understanding/`).

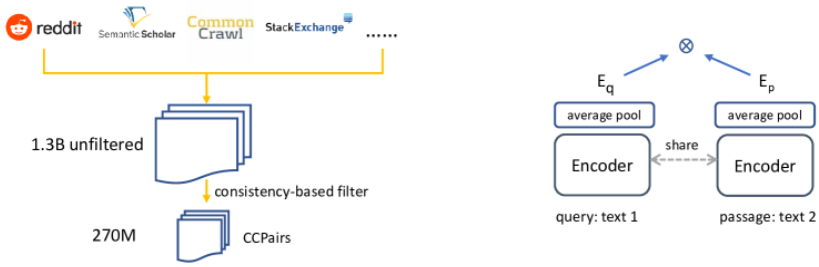


Figure 5: E5 two-stage training pipeline: weakly-supervised contrastive pre-training on CCPairs (Stage 1) followed by fine-tuning on labelled data (Stage 2). Source: Wang et al. (2022) via ar5iv (`https://arxiv.org/abs/2212.03533`).

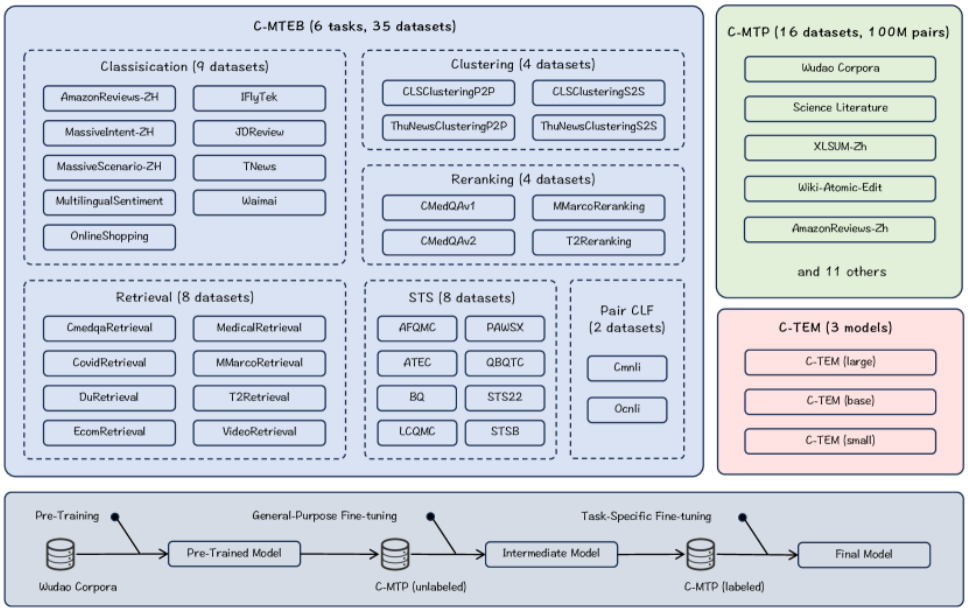


Figure 6: BGE/C-Pack training recipe: large-scale contrastive pre-training followed by hard-negative mining and instruction-aware fine-tuning. Source: Xiao et al. (2023) via ar5iv (`https://arxiv.org/abs/2309.07597`).

6.3 Experimental Results

Table 10: Dense retrieval: NDCG@10. Bold = best dense model per dataset. DPR included as a cautionary baseline.

Model	Params	TREC-COVID	FIQA	SciFact
DPR	110M	0.1444	0.0601	0.2189
MiniLM-L6	22M	0.4725	0.3687	0.6451
MPNet-base	110M	0.5133	**0.4996**	0.6557
E5-base	110M	0.6963	0.3987	0.7194
BGE-base	110M	**0.7807**	0.4062	**0.7404**
BM25 (reference)	N/A	0.4471	0.1591	0.5597

Findings. BGE dominates on TREC-COVID and SciFact; instruction-tuned contrastive training with hard-negative mining gives it a clear edge. MPNet is the best model on FIQA despite being older and same-sized as BGE and E5; its permuted language model objective may better align with conversational financial queries. DPR is a cautionary result: NQ-only training collapses on all three out-of-domain benchmarks, performing below BM25 and even below Word2Vec on TREC-COVID. All general sentence encoders surpass BM25 on all three datasets; model breadth of training distribution matters more than domain-specific fine-tuning for zero-shot generalisation.

Illustrative per-query examples.

- **DPR failure (domain shift).**
  DPR scores NDCG@10 0.060 on FIQA and 0.219 on SciFact — below BM25 and

even Word2Vec on all three benchmarks. Trained exclusively on Natural Questions (Wikipedia QA), DPR is not architecturally broken; it is the narrowest training distribution of any model evaluated and collapses completely outside its training domain.

- **All dense models solved the OOV problem that Word2Vec could not.**
  Query: "ADAR1 binds to Dicer to cleave pre-miRNA." (SciFact)
  Expected passage: ADAR1 Forms a Complex with Dicer to Promote MicroRNA Processing. "ADAR1 forms a complex with Dicer through direct protein-protein interaction and increases the maximum rate of pre-miRNA cleavage."
  Why it worked: All transformer-based models (MiniLM, MPNet, BGE) use subword tokenisation, decomposing `ADAR1` and `Dicer` into known byte-pair units. All three rank the relevant passage at position 1. Word2Vec dropped both identifiers silently as OOV and retrieved an unrelated passage.

- **BGE succeeded where other dense models struggled (hard-negative training).**
  Query: "Ubiquitin ligase UBC13 generates a K63-linked polyubiquitin moiety at PCNA K164." (SciFact)
  Expected passage: Replication Fork Slowing and Reversal upon DNA Damage Require PCNA Polyubiquitination. "DNA damage tolerance during eukaryotic replication is orchestrated by PCNA ubiquitination; polyubiquitination activates an error-free damage bypass pathway."
  Retrieved (rank 1) by MiniLM and MPNet: Evolution and function of ubiquitin-like protein-conjugation systems. (related ubiquitin topic but not the specific PCNA K164 pathway; first relevant at ranks 10 and 5 respectively)
  Why BGE won: Hard-negative mining during training forces BGE to distinguish between passages from the same topic area (ubiquitin biology). MiniLM and MPNet, trained without hard negatives, conflate the general ubiquitin-pathway passage with the specific PCNA K164 claim.

- **Dense models partially failed (clinical specificity bottleneck).**
  Query: "p16INK4A accumulation is linked to an abnormal wound response caused by the microinvasive step of advanced Oral Potentially Malignant Lesions (OPMLs)." (SciFact)
  Expected passage: Monitoring Tumorigenesis and Senescence In Vivo with a p16INK4a-Luciferase Model.
  Retrieved (rank 1): MiniLM and BGE retrieve *p16INK4A is a robust in vivo biomarker of cellular aging* (related but wrong claim); MPNet retrieves *Tumor-associated macrophages correlate with EMT in oral squamous cell carcinoma* (shares `oral` + cancer context).
  Why it partially failed: MiniLM and BGE retrieve the relevant passage within the top-10 (ranks 3 and 4) but not at rank 1; MPNet misses it entirely (rank 13, NDCG@10 = 0.000). The three clinical identifiers (`p16INK4A`, `microinvasive`, `OPMLs`) introduce enough specificity that even subword tokenisation cannot fully preserve their combined meaning in a single fixed-size vector.

7 Phase 4: Hybrid Retrieval and Reranking

7.1 Why Combine Lexical and Dense Retrieval?

Sparse and dense retrievers have complementary strengths and weaknesses:

- **BM25** excels at *exact-match* for rare, discriminative terms (acronyms, named entities, product IDs, molecular names). It fails when the query and relevant passage use different vocabulary (synonym gap).
- **Dense models** excel at *semantic matching*; they retrieve passages that are conceptually similar even with zero term overlap. They can struggle with very specific rare terms that were unseen or rare in pre-training.

RRF fusion (Equation 5, $k = 60$) combines both ranked lists without requiring a trained combiner, making it parameter-free and robust. The $k = 60$ default prevents a single high-rank appearance in one list from dominating the fused score.

7.2 System Architecture

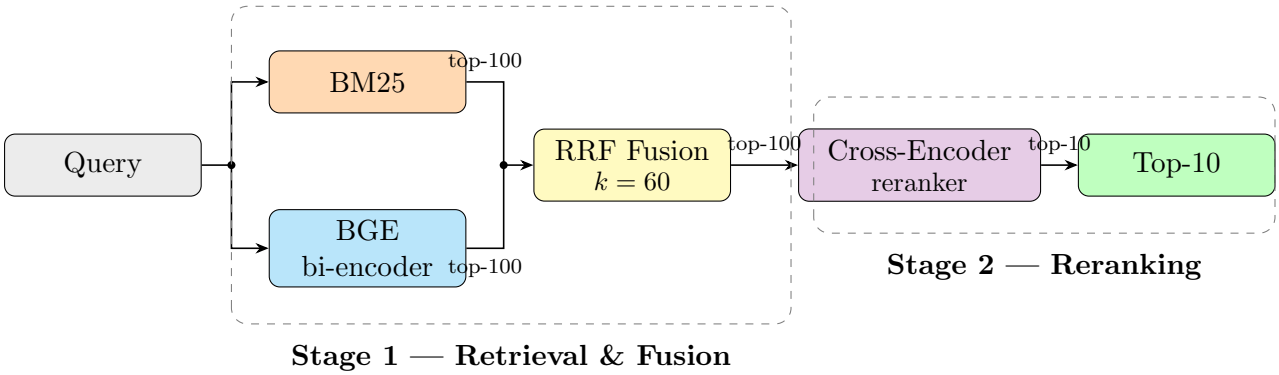


Figure 7: Two-stage hybrid retrieval pipeline. **Stage 1 (Retrieval & Fusion):** BM25 and BGE independently retrieve the top-100 candidates; their ranked lists are merged via Reciprocal Rank Fusion ($k = 60$) into a unified top-100 candidate set. **Stage 2 (Reranking):** a cross-encoder (`ms-marco-MiniLM-L-6-v2`) scores every (query, passage) pair in the candidate set and returns the final top-10.

1. BM25 and BGE independently retrieve top-100 candidates.
2. Scores are fused via RRF (Equation 5, $k = 60$).
3. A cross-encoder (`cross-encoder/ms-marco-MiniLM-L-6-v2`) reranks the top-100 RRF candidates.
4. Top-10 are returned for evaluation.

Cross-encoder vs. bi-encoder. The cross-encoder processes the concatenated sequence [CLS] q [SEP] p [SEP] through a single transformer, allowing every query token to attend to every passage token:

$$\text{score}_{\text{CE}}(q, p) = \text{Linear}\big(\text{BERT}([\text{CLS}]\ q\ [\text{SEP}]\ p)_{[\text{CLS}]}\big) \in \mathbb{R} \tag{20}$$

This *full interaction* produces more accurate relevance scores than a dot product over independent vectors. The trade-off: passage embeddings cannot be pre-computed, so inference cost is $O(|Q| \times |\text{candidates}|)$, making it practical only for reranking a small candidate set (top-100).

7.3 Experimental Results

Table 11: Hybrid retrieval: NDCG@10. Bold = best overall per dataset.

Method	TREC-COVID	FIQA	SciFact
BM25 (sparse)	0.4471	0.1591	0.5597
BGE (best dense)	**0.7807**	**0.4062**	**0.7404**
Hybrid RRF	0.7100	0.2921	0.6674
Hybrid + Reranker	0.7630	0.3743	0.6888

Table 12: Recall and MAP for hybrid methods across all datasets.

Method	MAP@100	Recall@10	Recall@50	Recall@100
FIQA				
Hybrid RRF	0.2423	0.3690	0.6255	0.6964
Hybrid + Reranker	0.3119	0.4574	0.6334	0.6965
SciFact				
Hybrid RRF	0.6296	0.7977	0.9343	0.9527
Hybrid + Reranker	**0.6520**	**0.8089**	0.9303	0.9527
TREC-COVID				
Hybrid RRF	0.0830	0.0189	0.0723	0.1197
Hybrid + Reranker	0.0932	0.0209	0.0837	0.1195

Findings. Counter to the common assumption that hybrid always beats dense, **BGE alone outperforms Hybrid+Reranker** on both TREC-COVID (0.781 vs. 0.763) and FIQA (0.406 vs. 0.374). The culprit is BM25: when the sparse component is weak (conversational queries on FIQA; highly specialised biomedical vocabulary on TREC-COVID), RRF fusion *degrades* the dense signal. Adding the cross-encoder reranker partially recovers this loss (+0.05 on TREC-COVID, +0.08 on FIQA), but does not reach BGE alone on TREC-COVID.

The reranker does not improve Recall@100; it only reorders the same 100 candidates, and Recall@100 matches exactly for RRF vs. Reranker. The reranker's value is better precision at small k: Recall@10 improves by +0.088 on FIQA and +0.011 on SciFact.

Illustrative per-query examples.

- **RRF hurt on SciFact.** BGE alone achieves Recall@100 = 0.967 on SciFact, already covering 96.7% of relevant passages. Equal-weight RRF allows BM25's top-ranked (often incorrect) passages to displace BGE's lower-ranked but still relevant ones, reducing NDCG@10 from 0.740 to 0.667 (−0.073). When one retriever dominates, equal-weight fusion dilutes rather than complements.
- **Cross-encoder recovered precision on FIQA.** The reranker improves Recall@10 from 0.369 to 0.457 (+8.8 pp) by jointly encoding each (query, passage) pair. A bi-encoder dot product scores query and passage independently, so it can miss fine-grained distinctions between passages that share surface vocabulary but address different questions. The cross-encoder attends across both texts simultaneously, providing a more accurate relevance signal within the re-ranked shortlist.

8 Extension A: Domain-Specific Fine-tuning

8.1 Setup

We fine-tune `all-MiniLM-L6-v2` (22M parameters) using MultipleNegativesRankingLoss (Equation 8). Training pairs are constructed from BEIR qrels: each (query, positive passage) pair from the training split. Three fine-tuning variants are evaluated.

Table 13: Fine-tuning experimental variants.

Variant	Train pairs	Time	Epochs
ft-FIQA	14,131	6.3 min	3
ft-SciFact	919	1.1 min	5
ft-Combined	15,050	6.7 min	3

8.2 Experimental Results

Table 14: Fine-tuning results: NDCG@10. Base model = `all-MiniLM-L6-v2`. Bold = improvement over base.

Model	Train pairs	TREC-COVID	FIQA	SciFact
MiniLM base	N/A	0.4725	0.3687	0.6451
ft-FIQA	14,131	**0.4867**	**0.3964**	0.6451
ft-SciFact	919	0.4606	0.3593	**0.6949**
ft-Combined	15,050	**0.5027**	**0.3997**	**0.6596**

Findings.

- **In-domain specialisation is strong and data-efficient (Table 15).** Fine-tuning on 919 SciFact pairs (+7.7%) achieves comparable relative gain to 14,131 FIQA pairs (+7.5%). Recall@10 on SciFact jumps from 0.783 to 0.865 (+8.2 percentage points).

Table 15: Primary: in-domain gain. Each variant is judged on the dataset(s) it was trained on.

Fine-tuned on	Eval dataset	Base NDCG@10	Δ
SciFact	SciFact	0.6451	+0.0498
FIQA	FIQA	0.3687	+0.0277
Combined	SciFact	0.6451	+0.0145
Combined	FIQA	0.3687	+0.0310

Table 16: Secondary: cross-domain impact (catastrophic forgetting check). Eval dataset was *not* in the training set.

Fine-tuned on	Eval dataset	Base NDCG@10	Δ
SciFact	FIQA	0.3687	−0.0094
SciFact	TREC-COVID	0.4725	−0.0119
FIQA	SciFact	0.6451	0.0000
FIQA	TREC-COVID	0.4725	+0.0142
Combined	TREC-COVID	0.4725	+0.0302

- **Negative transfer is real but small (Table 16).** SciFact fine-tuning hurts FIQA (−0.009) and TREC-COVID (−0.012); domain specialisation trades breadth for depth. The penalty is modest compared to the in-domain gain.
- **Combined training generalises to unseen domains.** ft-Combined is the only variant that improves all three datasets simultaneously, including TREC-COVID (+3.0%) which appears in neither training set. TREC-COVID has no training split in BEIR; ft-Combined is the closest proxy available, and its cross-domain transfer demonstrates that multi-domain training reduces specialisation cost.
- **Small model + in-domain data beats larger zero-shot models.** ft-SciFact MiniLM (22M, NDCG 0.695) exceeds MPNet (110M, 0.656) and approaches E5 (110M, 0.719) on SciFact, with 5× fewer parameters with targeted data.

Embedding-space effect. Before fine-tuning, base MiniLM assigns irrelevant pairs a mean cosine similarity of 0.089 — uncomfortably close to the relevant-pair mean of 0.586. After fine-tuning on SciFact, irrelevant-pair similarity drops to 0.026 while the relevant-pair mean holds at 0.573. The separation gap widens from 0.497 to 0.547: the gain does not come from pulling relevant passages closer, but from pushing irrelevant ones further away. This directly explains the +7.7% NDCG improvement.

Illustrative per-query improvement.

- **Domain-specific claim matched after fine-tuning.** *Query:* "The PPR MDA5 has two N-terminal CARD domains." *Expected passage (title):* "Immune signaling by RIG-I-like receptors."
  Base MiniLM (rank 161): Returns "Chaperoning 5S RNA assembly" — a ribosomal RNA paper with no connection to innate immunity. The base model has no representation of *MDA5* (a pattern recognition receptor) or *CARD domains* in the antiviral

signalling context, so it makes an arbitrary match on surface-level RNA co-occurrence. *Fine-tuned MiniLM (rank 2):* The model trained on SciFact biomedical claim–evidence pairs learns that claims about specific protein domain architecture (CARD, DEAD-box helicase) belong in the innate immunity literature. The correct review of RIG-I-like receptors — which explicitly covers MDA5 domain structure — advances from rank 161 to rank 2.

9 Note on LLM-as-Judge Evaluation

Standard IR evaluation relies on manually curated qrels. For all experiments in this study, NDCG and MRR are the right metrics: the BEIR benchmarks provide high-quality human relevance judgements, and qrel-based metrics are reproducible, fast, and interpretable.

LLM-as-judge evaluation (prompting a language model to score each retrieved passage for relevance) is a separate technique warranting consideration in the following scenarios.

When LLM-as-judge adds value.

- **No ground-truth labels.** For a private corpus or a domain not covered by existing benchmarks, there are no qrels. An LLM judge provides the only automated relevance signal without costly human annotation.
- **Sparse qrel coverage (annotation gaps).** Benchmark qrels are constructed by pooling candidates from a baseline retriever. A stronger model may surface genuinely relevant passages that were never judged; these appear as false positives in NDCG even when they are correct. An LLM judge can identify such cases.
- **End-to-end RAG evaluation.** Once a generator is in the pipeline, the question shifts from "did the right chunk rank first?" to "did the generated answer use the retrieved evidence correctly?" Metrics such as faithfulness and answer relevancy require an LLM to assess; there is no qrel-based equivalent.
- **Fine-tuned model on held-out queries without annotated qrels.** If the deployment domain has no labelled test split, an LLM judge provides a directional signal for measuring improvement.

Key caveats.

- **The judge must be capable for the domain.** A small model (e.g. 3B parameters) judging specialised scientific claims will produce unreliable scores , not because the technique is broken, but because the model lacks the domain knowledge to assess relevance accurately. For specialist domains, a larger or domain-adapted judge is required.
- **Scores are not comparable across judge models.** Absolute context-precision values are specific to the judge setup; only relative comparisons within the same configuration are meaningful.
- **Cost scales with queries $\times$ candidates.** Each (query, passage) pair requires one LLM call. Sample 100–200 queries for a directional signal rather than scoring the full set.

For the retrieval paradigms studied here (sparse, dense, hybrid, and fine-tuned bi-encoders on established BEIR benchmarks), NDCG@10 is the appropriate primary metric throughout.

10 Discussion

10.1 When Does Hybrid Retrieval Help?

Hybrid RRF improves over both components only when both are competitive. When BM25 is systematically weak (conversational or highly domain-specific queries), it introduces noise into the fusion and RRF can *hurt* dense retrieval. The cross-encoder reranker is a more robust upgrade: it acts on the candidate set regardless of how it was assembled, and consistently yields +0.02–0.08 NDCG across all datasets.

Practical implication: for production RAG systems, prefer dense-only + cross-encoder over BM25+dense hybrid unless BM25 quality on the target domain is validated first.

10.2 Fine-tuning Efficiency vs. Scale

The SciFact experiment demonstrates that *very few* high-quality training pairs suffice for significant improvement (919 pairs, 1.1 min on Apple M5 MPS). The key mechanism is MNR loss's in-batch negatives: with batch size 64, each positive pair is contrasted against 63 negatives per step without requiring hard-negative mining.

10.3 When to Move Beyond NDCG

NDCG is the right metric when high-quality qrels exist, as in all three benchmarks used here. The main scenario where it falls short is *annotation gap*: a better model retrieves genuinely relevant passages that pooling-based annotation never assessed, and NDCG penalises them as false positives.

For production systems without labelled data, LLM-as-judge provides a directional signal, but the judge must match the domain. A small general model judging scientific claims or biomedical queries will apply generic topical relevance rather than the task-specific relevance criteria that the qrels encode. The practical implication: use a capable, domain-appropriate judge, treat scores as relative rather than absolute, and reserve the technique for the scenarios listed in Section 9.

11 Conclusion

We have traced the evolution of information retrieval across four paradigms, from BoW to fine-tuned neural encoders, on three heterogeneous BEIR benchmarks. Key takeaways:

1. BM25 is a strong, zero-cost baseline but *not always the best sparse method*: TF-IDF's implicit global normalisation outperforms BM25 on short, uniform corpora (SciFact, FIQA).

2. Dense bi-encoders with general contrastive training (BGE, E5) beat BM25 across all three datasets. Task-specific fine-tuning (DPR) without data diversity produces

brittle, domain-fragile models, performing below BM25 and Word2Vec on out-of-domain benchmarks.

3. Hybrid RRF is not universally better than dense retrieval; it requires a competitive sparse component. The cross-encoder reranker is a more reliable upgrade (+0.02–0.08 NDCG across all settings).

4. Domain-specific fine-tuning with MNR loss is highly data-efficient; 919 SciFact pairs deliver +7.7% in-domain NDCG with under 2 minutes of training on consumer hardware. Combined multi-domain training generalises to unseen domains without negative transfer.

5. LLM-as-judge is not appropriate as a primary metric when annotated qrels exist. Its value is in evaluating private corpora with no labels, surfacing annotation gaps in large benchmarks, and assessing end-to-end RAG quality beyond rank-based metrics.

Acknowledgements

Code and Data

The full source for this study is available at
`https://github.com/saikrishnab-sahajai/retrieval-evolution-study`.
The repository includes all evaluation scripts, result files (JSON), and notebooks. Large corpus files and fine-tuned model checkpoints are excluded but can be reproduced via `scripts/download_datasets.py` and `scripts/run_finetune.py`.

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A Complete Metric Tables

Table 17: Full sparse retrieval metrics.

Dataset	**Method**	**NDCG@10**	**MRR@10**	**MAP@100**	**Recall@10**
TREC-COVID	BoW	0.1763	0.3177	0.0084	0.0035
	TF-IDF	0.2866	0.5025	0.0264	0.0077
	BM25	0.4471	0.7197	0.0413	0.0113
FIQA	BoW	0.0656	0.0889	0.0524	0.0791
	TF-IDF	0.1793	0.2156	0.1412	0.2435
	BM25	0.1591	0.1985	0.1239	0.2037
SciFact	BoW	0.3647	0.3424	0.3394	0.4617
	TF-IDF	0.6286	0.5897	0.5827	0.7735
	BM25	0.5597	0.5242	0.5202	0.6862

Table 18: Full dense retrieval metrics (NDCG@10, MRR@10, and Recall@10).

Dataset	**Model**	**NDCG@10**	**MRR@10**	**Recall@10**
TREC-COVID	MiniLM	0.4725	0.7275	0.0128
	MPNet	0.5133	0.7576	0.0147
	E5	0.6963	0.9133	0.0196
	BGE	0.7807	0.9180	0.0221
FIQA	MiniLM	0.3687	0.4543	0.4413
	MPNet	0.4996	0.5886	0.5811
	E5	0.3987	0.4862	0.4705
	BGE	0.4062	0.4931	0.4814
SciFact	MiniLM	0.6451	0.6110	0.7833
	MPNet	0.6557	0.6237	0.7901
	E5	0.7194	0.6880	0.8465
	BGE	0.7404	0.7071	0.8742